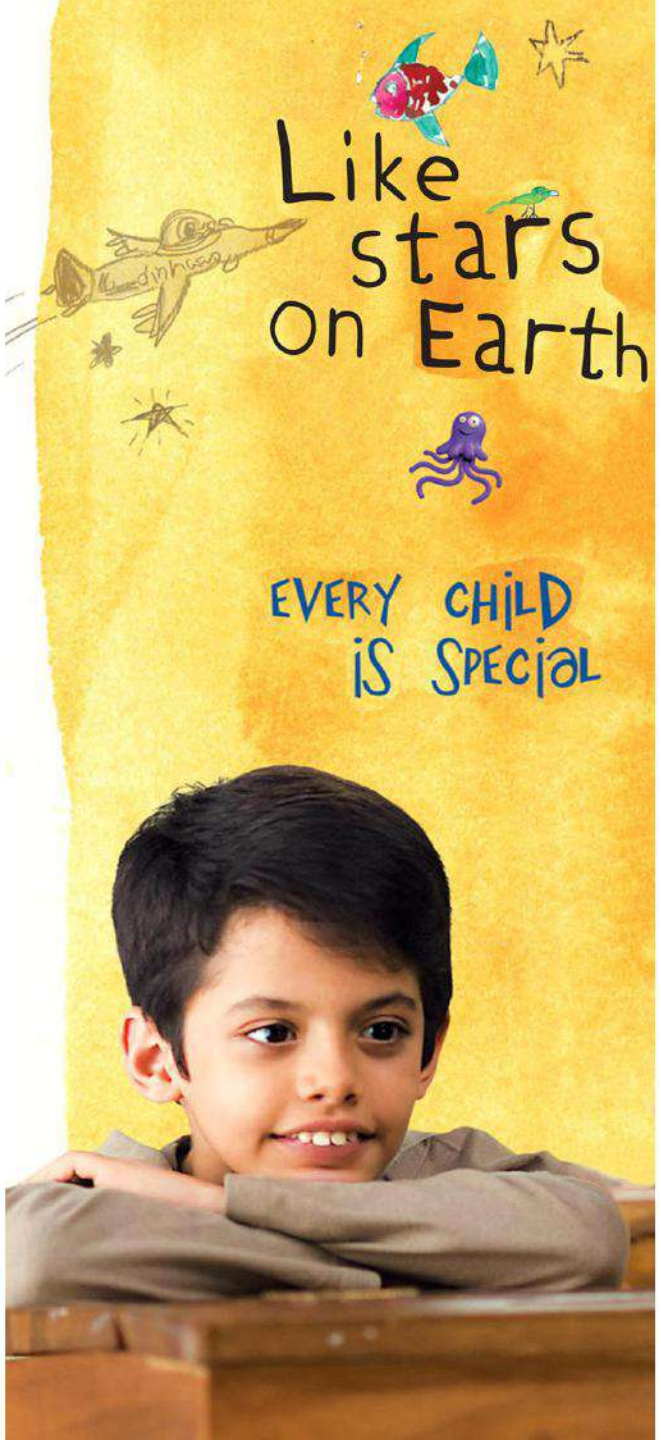


SCREENING AND REFINEMENT SYSTEM TO IDENTIFY THE RISK OF DYSCALCULIA AND DYSGRAPHIA LEARNING DISABILITIES IN PRIMARY SCHOOL STUDENTS

20_21-J 01

Members



Supervisor

Prof. Samantha Thelijjagoda



Co -supervisor

Ms. Jenny Krishara



External supervisor

Dr. W.M.K. Wanigasingha



Group Leader

Hewapathirana C.D.D.



Abeysinghe A.D.K.S.



Maheshani H.M.P.

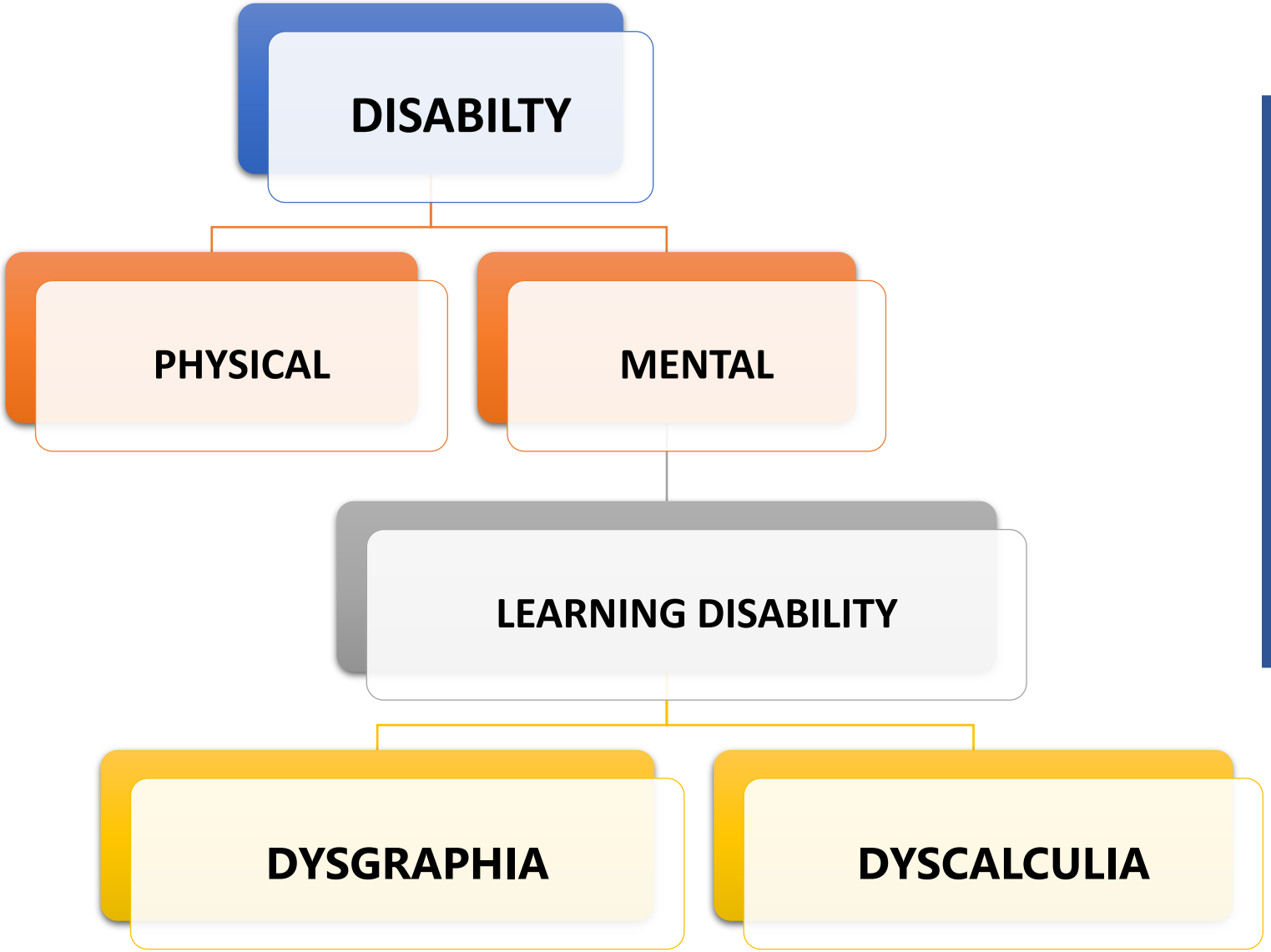


Liyanage T.J.P.S.

Content

1. Introduction
2. Research Question
3. Objectives
4. System Overview Diagram
5. Technologies
6. Individual Progress
7. Standards and Best practices
8. Requirements
9. Commercialization Aspects
10. Budget

1. INTRODUCTION





INTRODUCTION CONT.



NANA SHILPA

DYSGRAPHIA

Difficulty of writing in provided space



Unclear and irregular handwriting



Slow writing pace



DYSCALCULIA



Mathematics disorder



Difficulty in identifying numbers



Difficulty in perform basic mathematical operations

2. RESEARCH QUESTION

How to provide mobile application-based solution to

- Screen and predict the risk of Dyscalculia and Dysgraphia learning disabilities(LD) in primary school students ?
- Deliver Comprehensive activities to overcome those LDs risks?





3. OBJECTIVES

MAIN OBJECTIVE

- Provide mobile application-based solution to screen and predict the risk of Dyscalculia and Dysgraphia learning disabilities in primary school students.

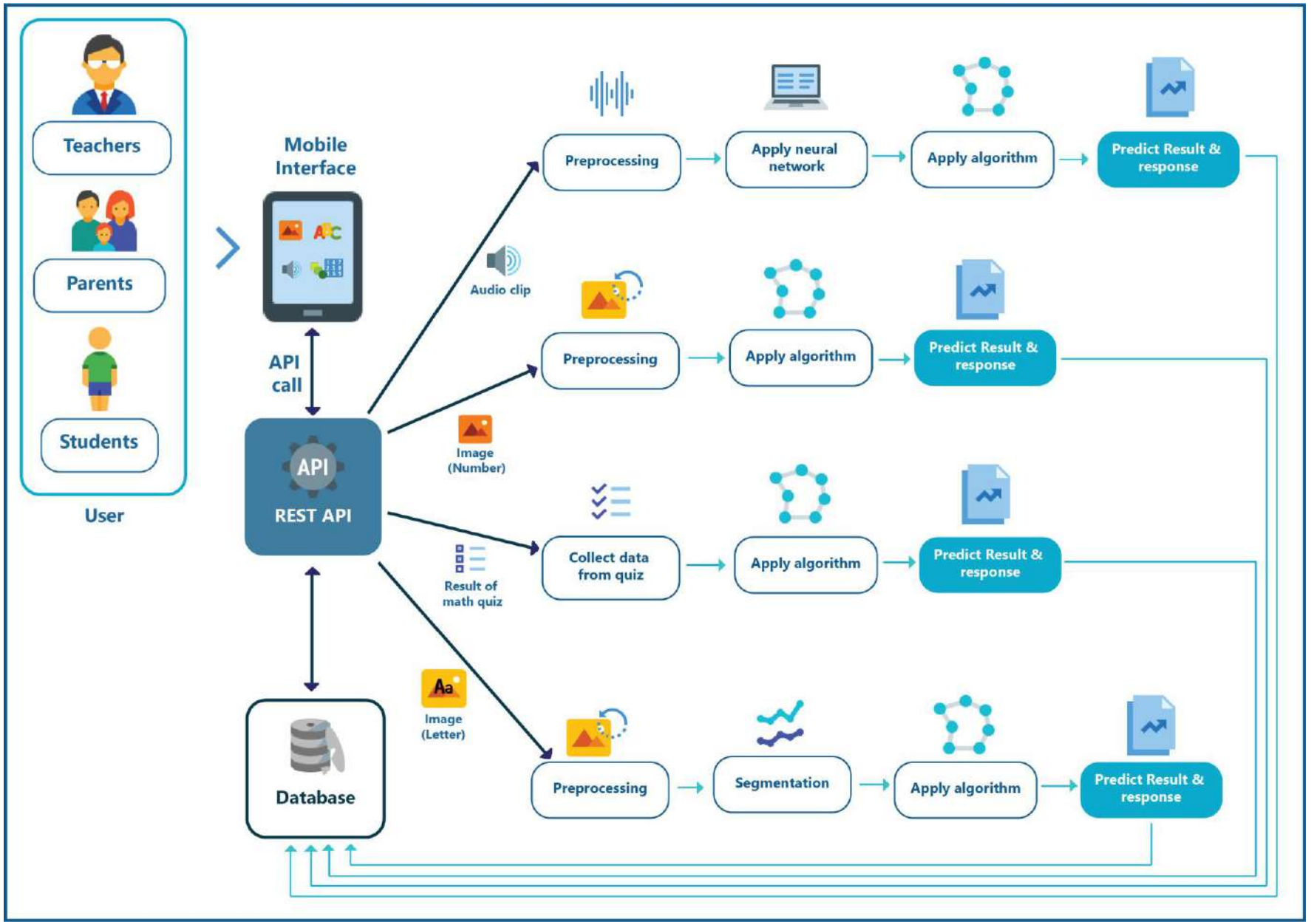


SPECIFIC OBJECTIVES



- Identify the risk of Letter level Dysgraphia through Sinhala Characters and provide activities to improve the condition.
- Identify the risk for Numeric Dysgraphia and provide activities to improve the condition.
- Identify the risk for Operational and Practognostic Dyscalculia and provide activities to improve the condition.
- Identify the risk for Lexical and Verbal Dyscalculia and provide activities to improve the condition.

4. SYSTEM OVERVIEW DIAGRAM



5. TECHNOLOGIES



Mobile Application

- Android

Database

- SQLite

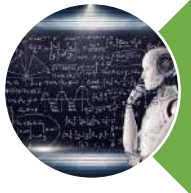
Middle Ware Technologies

- Python
- TensorFlow
- REST API - Flask



Technical Concepts

- Machine Learning
- Convolutional Neural Network (CNN)
- Image Processing



Algorithms

- Support Vector Machine
- FFT Algorithm



IT17180108 | Kalpani Sandupaba Abeysinghe A.D.

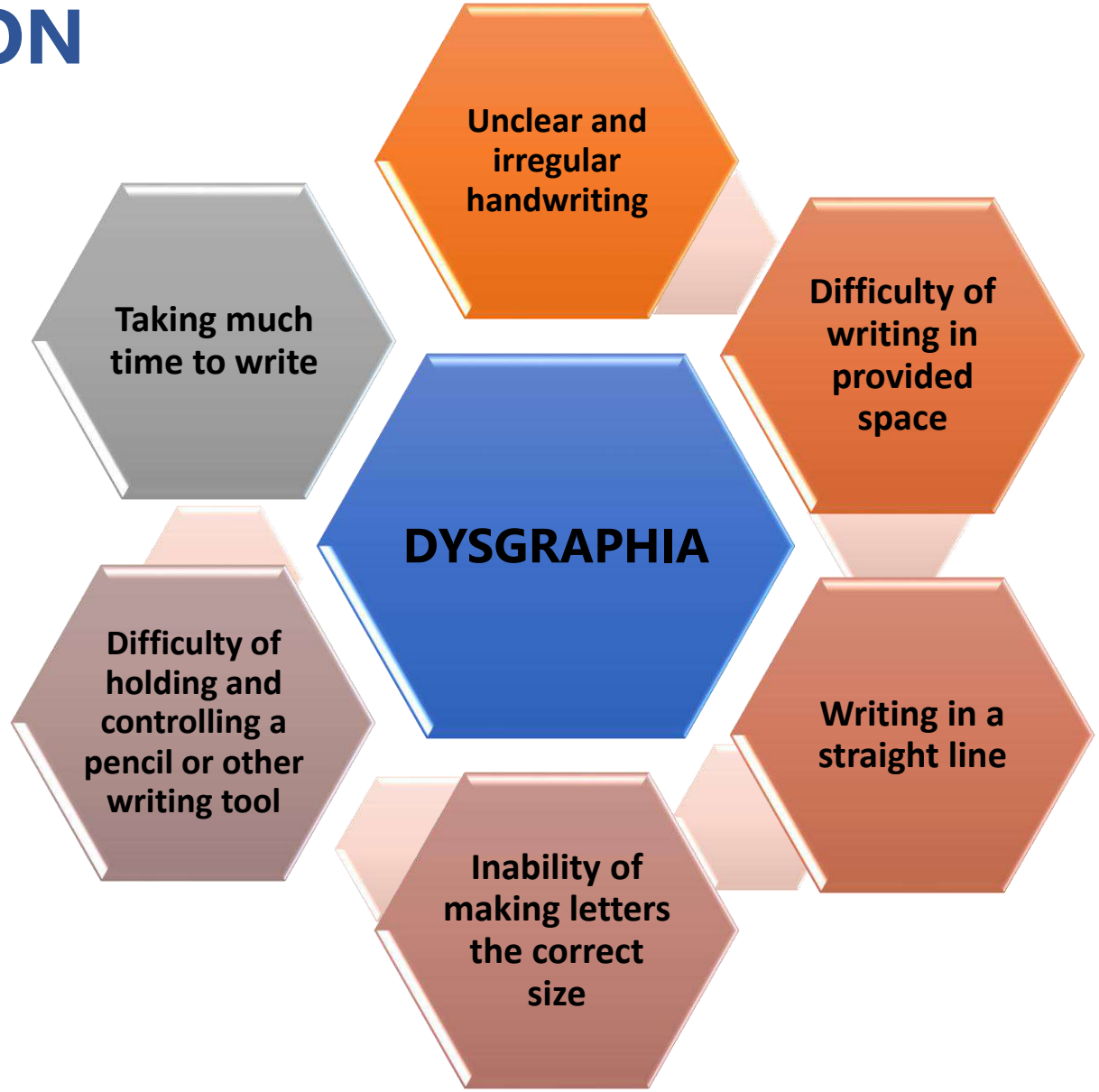
BSc (Hons) Degree in Information Technology (specialization in Software Engineering)

THE PROGRESS

Identify the risk of Letter level Dysgraphia through Sinhala Characters and provide activities to improve the condition

❖ INTRODUCTION

❖ BACKGROUND



❖ RESEARCH GAP

Assistive Learning Environment for Students with Learning Disabilities[3]

- Proposed Assistive Learning Environment (ALE) for the students who are undergoing with learning disabilities.
- Provide environment for the students in order to follow the activities in their own pace.

A Mobile Web Game Approach for Improving Dysgraphia[4]

- Provide game-based solution for dysgraphia students.
- Implemented using client server web application architecture.

An Expert System for Diagnosing Dysgraphia [5]

- Identify the Dysgraphia condition through the decision tree analyze techniques.

Automatic segmentation as a tool for examining the handwriting process of children with Dysgraphic and proficient handwriting [6]

- System for the teachers and students where teacher can view all the class through on interface.
- Based on the initial conducted activities, improvement learning model is assigned by the teacher.

❖ SYSTEM COMPARISON

Features	01	02	03	04	Nena Shilpa
Screening and intervention for Dysgraphia	✗	✗	✗	✗	✓
Letter Identification	✗	✗	✗	✗	✓
Supports native Sinhala language	✗	✗	✗	✗	✓
Recover activities based on the risk level	✗	✗	✗	✓	✓
Keep track of the progress	✗	✗	✗	✗	✓
Report Generation	✗	✗	✗	✗	✓
User attractive interfaces	✗	✓	✗	✓	✓

01-Assistive Learning Environment for Students with Learning Disabilities[3]
02 -A Mobile Web Game Approach for Improving Dysgraphia[4]
03 -An Expert System for Diagnosing Dysgraphia [5]
04 -Automatic segmentation as a tool for examining the handwriting process of children with Dysgraphic and proficient handwriting [6]

❖ RESEARCH PROBLEM

How to provide mobile application-based solution to

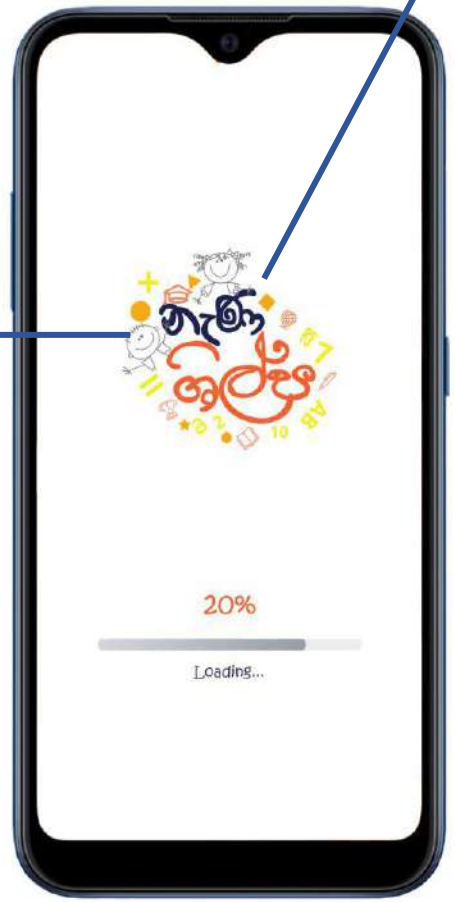
- **Identify the risk of letter level Dysgraphia condition ?**
- **Deliver risk diminish activities ?**
- **Monitor the betterment of the students ?**



❖ OBJECTIVES

SPECIFIC OBJECTIVE

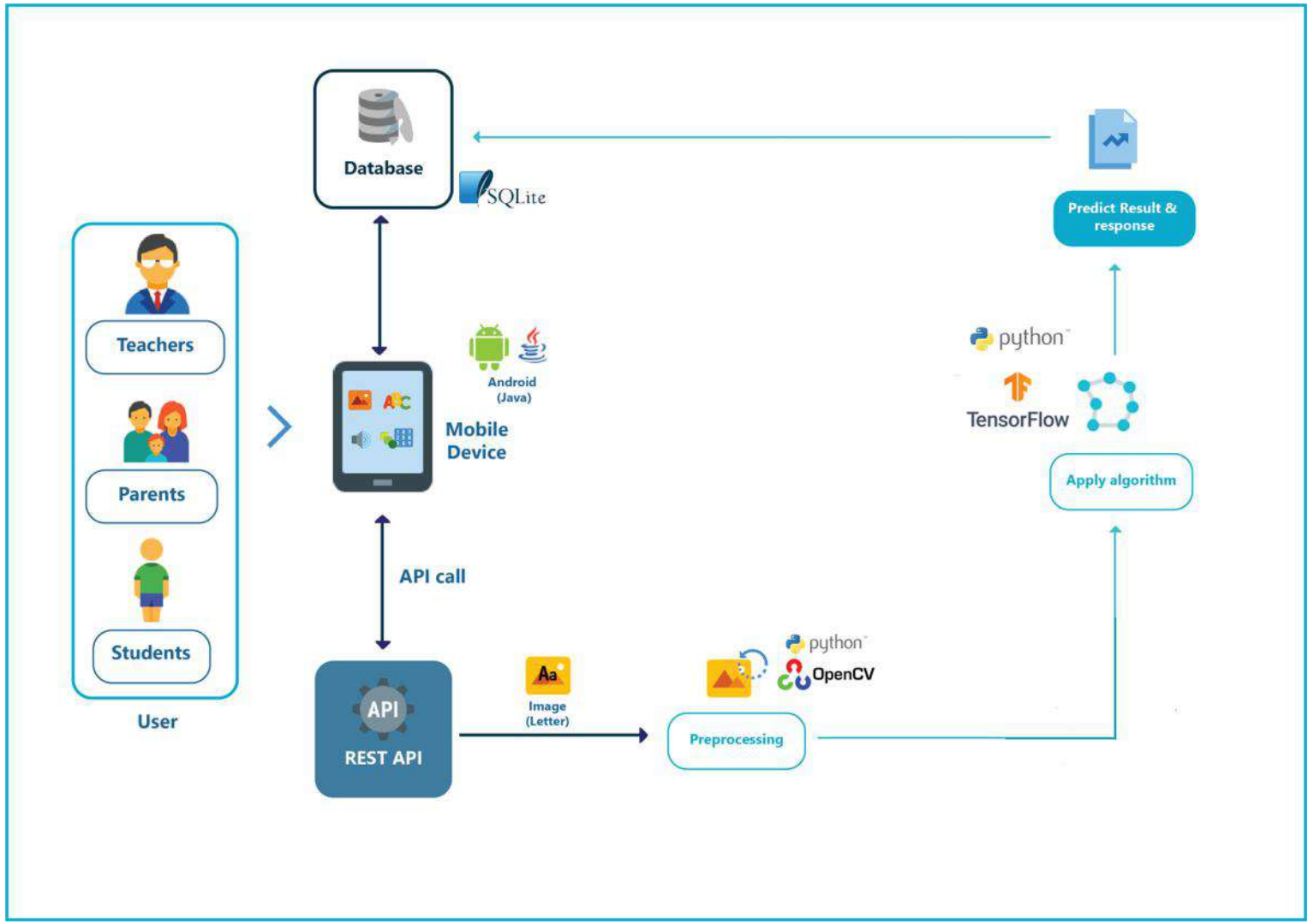
❖ Identify the risk of Letter level Dysgraphia through Sinhala Characters and provide activities to improve the condition.



Sub Objectives

- Letter level Dysgraphia risk identification.
- Practice activities to improve the condition.
- Keep the track of student activities.
- Report generation for the performed test activities.

METHODOLOGY

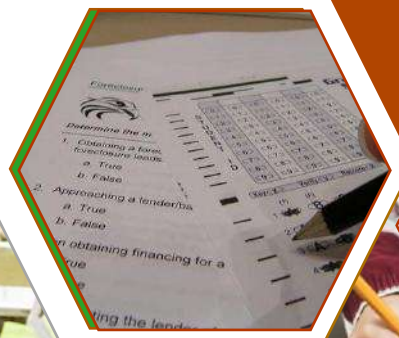


Component Overview Diagram

❖ KEY PILLARS

Optical Character Recognition

Trained Data Model







Machine Learning





Conventional Neural Network

Image Processing

❖ ACHIEVEMENTS



- Successful Data Collection for Model Training
- Completed Primary UI Design and Implementation
- User Input Collection Mechanism Implementation
- ML model has been trained with Accuracy 99%
- Backend Endpoint Implementation and Response Handling

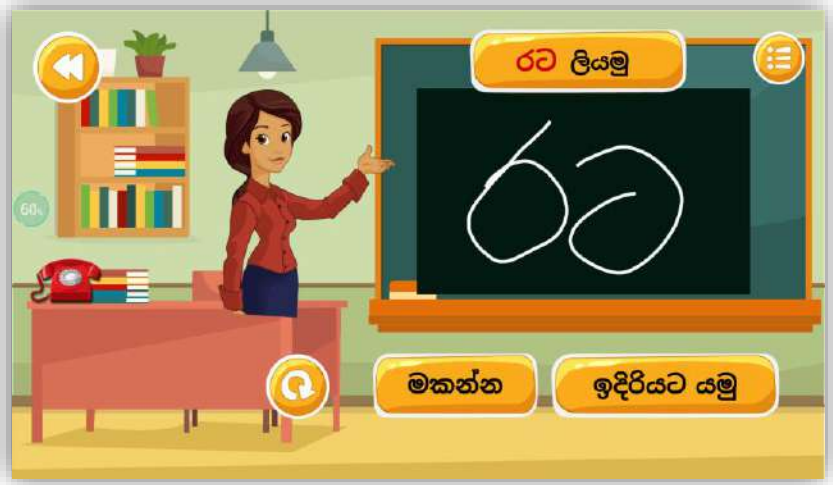
❖ ACHIEVEMENT EVIDENCE



Sub Menu



Screening UI for Letter



Screening UI for Word

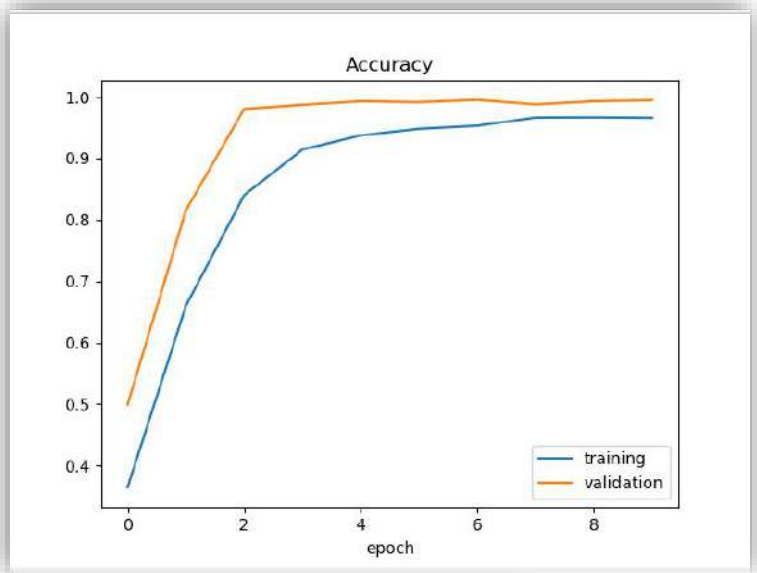


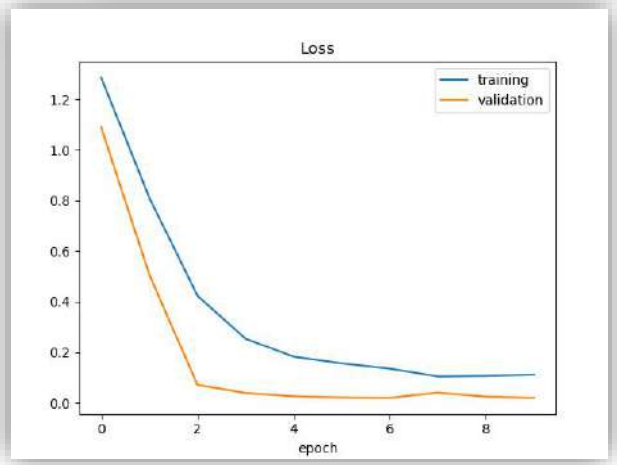
Sub Menu for Letter Intervention

❖ ACHIEVEMENT EVIDENCE CONTD.

```
Epoch 1/10
109/109 [==============================] - 16s 144ms/step - loss: 1.2859 - accuracy: 0.3645 - val_loss: 1.0893 - val_accuracy: 0.4993
Epoch 2/10
109/109 [==============================] - 16s 145ms/step - loss: 0.8111 - accuracy: 0.6597 - val_loss: 0.5072 - val_accuracy: 0.8151
Epoch 3/10
109/109 [==============================] - 14s 131ms/step - loss: 0.4237 - accuracy: 0.8398 - val_loss: 0.0716 - val_accuracy: 0.9803
Epoch 4/10
109/109 [==============================] - 15s 136ms/step - loss: 0.2542 - accuracy: 0.9140 - val_loss: 0.0392 - val_accuracy: 0.9868
Epoch 5/10
109/109 [==============================] - 15s 137ms/step - loss: 0.1824 - accuracy: 0.9371 - val_loss: 0.0260 - val_accuracy: 0.9927
Epoch 6/10
109/109 [==============================] - 15s 141ms/step - loss: 0.1563 - accuracy: 0.9478 - val_loss: 0.0213 - val_accuracy: 0.9912
Epoch 7/10
109/109 [==============================] - 16s 143ms/step - loss: 0.1354 - accuracy: 0.9528 - val_loss: 0.0200 - val_accuracy: 0.9949
Epoch 8/10
109/109 [==============================] - 16s 144ms/step - loss: 0.1047 - accuracy: 0.9660 - val_loss: 0.0407 - val_accuracy: 0.9876
Epoch 9/10
109/109 [==============================] - 16s 143ms/step - loss: 0.1067 - accuracy: 0.9664 - val_loss: 0.0247 - val_accuracy: 0.9927
Epoch 10/10
109/109 [==============================] - 16s 147ms/step - loss: 0.1115 - accuracy: 0.9657 - val_loss: 0.0200 - val_accuracy: 0.9942
Test Score =  0.02710871584713459
Test Accuracy = 0.9929783344268799

Process finished with exit code 0
```





CNN Model Accuracy

❖ WHAT IS TO BE DONE

Implement the Remedy Solution of the Component

Database Implementation for Report Management

Deploy the Backend to the Cloud Server

Integration of Backend and Frontend



❖ REFEREANCES

[01]Dysgraphia Information Page. (n.d.). Retrieved July 21, 2020, from https://www.ninds.nih.gov/Disorders/All-Disorders/Dysgraphia-Information-Page

[02]Mishna, F. (2003). Learning Disabilities and Bullying. *Journal of Learning Disabilities, 36*(4), 336-347. doi:10.1177/00222194030360040501

[03]Z. Pirani and M. Sasikumar, "Assistive learning environment for students with Learning Disabilities," *2012 4th International Conference on Intelligent Human Computer Interaction (IHCI)*, 2012

.

[04] D. Giordano and F. Maiorana, "A Mobile Web Game Approach for Improving Dysgraphia," Proceedings of the 7th International Conference on Computer Supported Education, 2015.

[05] D. A. Kurniawan, S. W. Sihwi, and Gunarhadi, "An expert system for diagnosing dysgraphia," 2017 2nd International conferences on Information Technology, Information Systems and Electrical Engineering (ICITISEE), 2017.

[06] Rosenblum, Sara & Dvorkin, Assaf & Weiss, Patrice. (2006). Automatic segmentation as a tool for examining the handwriting process of children with dysgraphic and proficient handwriting. Human movement science. 25. 608-21. 10.1016/j.humov.2006.07.005.



IT17182324 | Maheshani H.M.P

BSc (Hons) Degree in Information Technology (specialization in Software Engineering)

THE PROGRESS

Identify the risk for Numeric Dysgraphia and provide activities to improve the condition.

INTRODUCTION

❖ BACKGROUND

- A special Learning Disability called **Numeric Dysgraphia (ND)** not a very common term.
- This disability causes difficulties when writing the numbers.
- No special tests or treatments are not identified and cannot be cured [1].
- This kind of students can be identified by their numeric writing patterns by the teacher or a parent who pays much attention.



Through activities / practices, the situation of the student can be improved to some extent. So, it is important to identify this kind of students at early stage.

RESEARCH GAP

IO-CRIVO mobile app[1]

- For the students who has difficulties in writing.
- Screening for Dysgraphia
- PHP, HTML, CSS, JavaScript, MySQL

An Expert System[2]

- Provide remedy for Dysgraphic students on their studies
- A writing task and questionnaire was provided.

AR-DAWE model[3]

- AR based Dysgraphia Assistance Writing Environment (AR-DAWE).
- Identify spelling mistakes of the students.

A writing questionnaire[4]

- A statistical model
- A questionnaire

SYSTEM COMPARISON

Features	[1]	[2]	[3]	[4]	Nena Shilpa
Screening or intervention for Dysgraphia	✓	✓	✓	✓	✓
Screening and intervention for Dysgraphia	✗	✗	✗	✗	✓
Screening for Numeric Dysgraphia	✗	✗	✗	✗	✓
Remedies for ND in activity-based environment	✗	✗	✗	✗	✓
Keep track of the progress	✓	✗	✗	✗	✓
Support for Sinhala Language	✗	✗	✗	✗	✓
User attractive interfaces with sound bases	✗	✗	✓	✗	✓

RESEARCH PROBLEM

How to develop a solution which is a mobile application,



To identify the risk for ND ?



To provide remedies ?



To keep the track of progress ?



OBJECTIVES

SPECIFIC OBJECTIVE

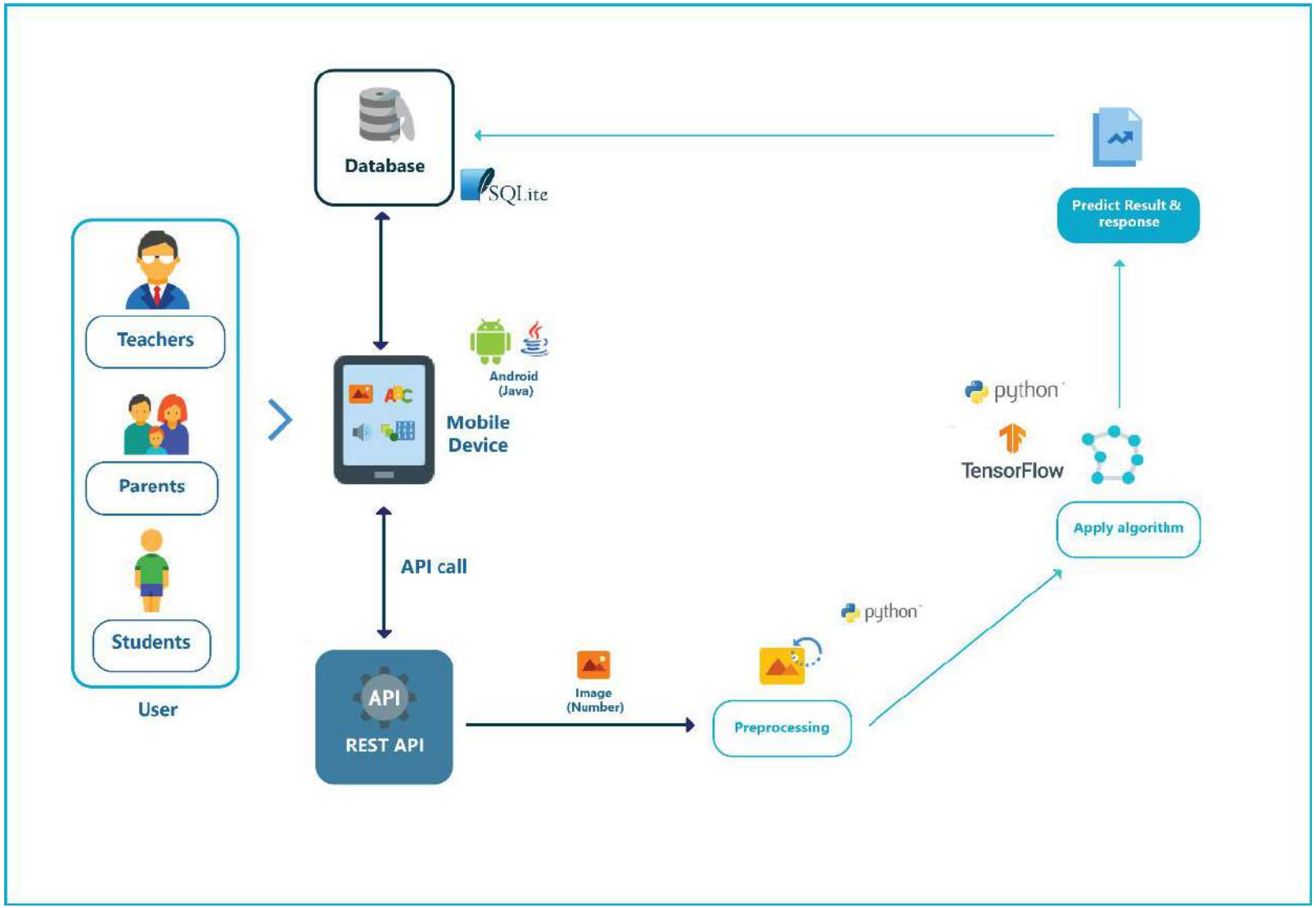
- Identify the risk for Numeric Dysgraphia and provide remedies while keeping the track of progress.



Sub Objectives

- Screening for the risk for Numeric Dysgraphia
- Providing activity-based treatments
- Keep the track of progress

METHODOLOGY



System Component Overview Diagram

KEY PILLARS



Machine Learning



Tensor Flow Lite

Random Forest Algorithm

Deep Learning

Convolutional Neural Network







❖ ACHIEVEMENTS



- Data gathered successfully.
- Kept the track of progress.
- Successfully trained the CNN 99% accuracy percentage.
- Achieved 99% accuracy percentage in ML model training.
- UI/UX designs

❖ ACHIEVEMENTS EVIDENCES



Screening UI



Current Quiz Result UI



Show Results Stages UI



Show Results Stages with details UI

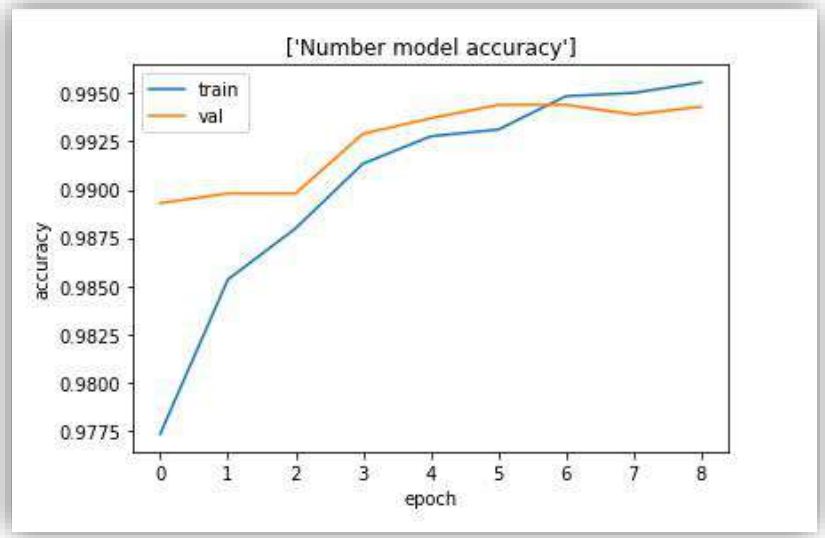
❖ ACHIEVEMENTS EVIDENCES Cont..

Model: "model_1"

Layer (type)	Output Shape	Param #
input (InputLayer)	[(None, 28, 28, 1)]	0
conv2d_3 (Conv2D)	(None, 23, 23, 24)	888
batch_normalization_4 (Batch	(None, 23, 23, 24)	72
activation_4 (Activation)	(None, 23, 23, 24)	0
dropout_4 (Dropout)	(None, 23, 23, 24)	0
conv2d_4 (Conv2D)	(None, 10, 10, 48)	28848
batch_normalization_5 (Batch	(None, 10, 10, 48)	144
activation_5 (Activation)	(None, 10, 10, 48)	0
dropout_5 (Dropout)	(None, 10, 10, 48)	0
conv2d_5 (Conv2D)	(None, 4, 4, 64)	49216
batch_normalization_6 (Batch	(None, 4, 4, 64)	192
activation_6 (Activation)	(None, 4, 4, 64)	0
dropout_6 (Dropout)	(None, 4, 4, 64)	0
flatten_1 (Flatten)	(None, 1024)	0
dense_1 (Dense)	(None, 200)	205000
batch_normalization_7 (Batch	(None, 200)	600
activation_7 (Activation)	(None, 200)	0
dropout_7 (Dropout)	(None, 200)	0
output (Dense)	(None, 10)	2010

Total params: 286,970
Trainable params: 286,298
Non-trainable params: 672

```
313/313 [==============================] - 4s 12ms/step - loss: 0.0171 - accuracy: 0.9943
Test error:  0.017147438600659 37, 'Test Accuracy: ' 0.9943000078201294
```



Trained CNN model Accuracy

❖ ACHIEVEMENTS EVIDENCES Cont..

```
]: experiment_accuracy = sm.accuracy_score(y_test, prediction_y)
   print('Accuracy score is : ', str(experiment_accuracy))

   Accuracy score is :  0.9896907216494846
```

Accuracy of Trained ML Model

```
Classification report:                    precision    recall  f1-score   support

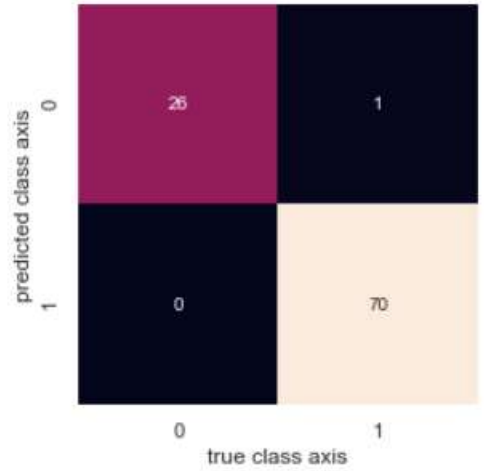
   Numeric Dysgraphia       1.00      0.96      0.98        27
No Numeric Dysgraphia       0.99      1.00      0.99        70

             accuracy                           0.99        97
            macro avg       0.99      0.98      0.99        97
         weighted avg       0.99      0.99      0.99        97
```

Classification Report of Trained ML Model

```
: sb.heatmap(confusionmt.T, square = True, annot = True, fmt = 'n', cbar = False)
  pt.xlabel('true class axis')
  pt.ylabel('predicted class axis')

: Text(89.18, 0.5, 'predicted class axis')
```



Show Test Results Using a Diagram

❖ WHAT IS TO BE DONE ??



- Integrate Frontend with Backend
- Intervention
- Deploy the Backend to the Cloud Server
- Finishing UI/UX designs

❖ REFERENCES

[1]. Giordano, D. & Maiorana, Francesco. (2014). Addressing dysgraphia with a mobile, web-based software with interactive feedback. 2014 IEEE-EMBS International Conference on Biomedical and Health Informatics, BHI 2014. 264-268. 10.1109/BHI.2014.6864354.

[2]. Kurniawan, Dimas & widya sihwi, Sari & Gunarhadi, Gunarhadi. (2017). An expert system for diagnosing dysgraphia. 468-472. 10.1109/ICITISEE.2017.8285552.

[3]. Khan, Muhammad & Hussain, Muhammad & Ahsan, Kamran & Saeed, Muhammad & Nadeem, Adnan & Ali, Syed & Mahmood, Nadeem & Rizwan, Kashif. (2017). Augmented Reality Based Spelling Assistance to Dysgraphia Students. Journal of Basic & Applied Sciences. 13. 500-507. 10.6000/1927-5129.2017.13.82.

[4]. Rosenblum, Sara & Dror, Gideon. (2016). Identifying Developmental Dysgraphia Characteristics Utilizing Handwriting Classification Methods. IEEE Transactions on Human-Machine Systems. PP. 1-7. 10.1109/THMS.2016.2628799.

[5]. 5 Best Machine Learning Algorithms for Classification Problems [online] Available: http://www.intellspot.com/machine-learning-classification/



IT17182638 | Chamil Diluksha Deshan

BSc (Hons) Degree in Information Technology (specialization in Software Engineering)

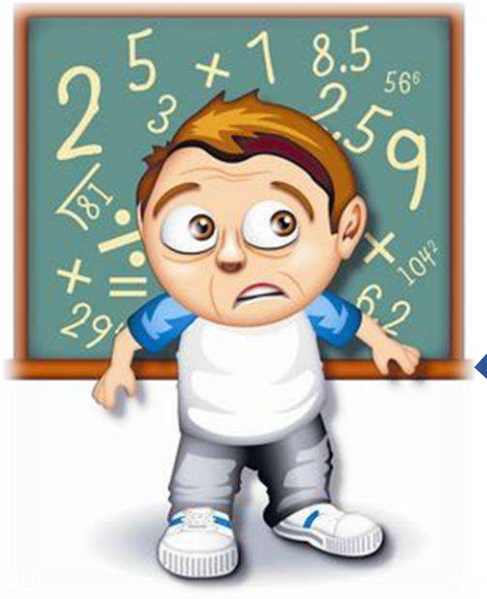
THE PROGRESS

Identify the risk for Operational and Practognostic Dyscalculia and provide activities to improve the condition.

INTRODUCTION

❖ Background

- ❖ Dyscalculia or mathematics learning disability it is a specific learning disability directed to arithmetic.[1][2].
- ❖ Dyscalculia is sometimes called as number blindness.
- ❖ The prevalence of developmental dyscalculia is between 3% and 6% in the school population.
- ❖ Among the dyscalculia types two of them are focusing on here,
  - **Operational Dyscalculia**
  - **Practognostic Dyscalculia**



RESEARCH GAP

Screening for Disorders of Mathematics via a web application [3]

- Learning process with calculation, numeric concepts, and problem solving.
- HTML5, CSS, JavaScript while using the MySQL as the database.

Development of Assistive Technology for Students with Dyscalculia [4]

- Provided Solution as a software tool which named as "Calculating Aid Tools: KidKanit" which helps to dyscalculia students in mathematical area.

Web-based learning tool for primary school student with Dyscalculia [5]

- Cover the most common characteristics which were the Dyscalculia students struggle with in specific areas such as counting, number recognition.

Screening for Dyscalculia within HE (Higher Education) [6]

- Provide screening process to identify dyscalculia in higher education students.
- In their solution, it was made a mobile app called DyscalculiUM.

SYSTEM COMPARISON

Features	[3]	[4]	[5]	[6]	Nena Shilpa
Dyscalculic Identification.	✗	✗	✗	✓	✓
Provide activities to improve the math concepts.	✓	✓	✓	✗	✓
Support for Sinhala Language	✗	✗	✗	✗	✓
User attractive interfaces with sound bases (multi-scenery)	✗	✓	✗	✗	✓
Keep track of progress	✗	✗	✗	✗	✓
Report Generation	✗	✗	✗	✗	✓
Supervised Learning(uses Machine Learning)	✗	✗	✗	✗	✓

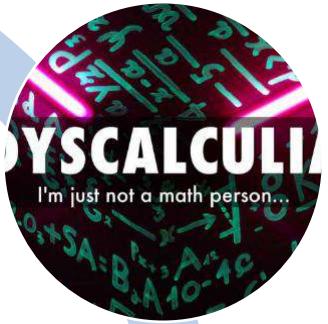




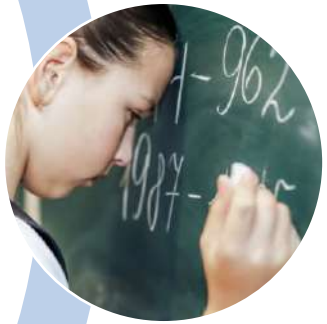
- Screening for Disorders of Mathematics via a web application [3]
- Development of Assistive Technology for Students with Dyscalculia [4]
- Web-based learning tool for primary school student with Dyscalculia [5]
- Screening for Dyscalculia within HE (Higher Education) [6]

RESEARCH PROBLEM

How to develop mobile **application-based solution to**



- Identify the risk for Operational and Practognostic Dyscalculia ?



- Provide activities to improve the condition ?



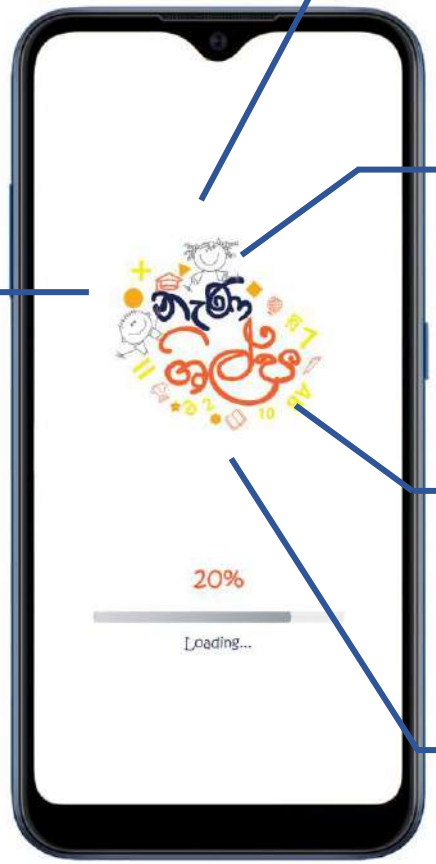
- Keep the track of progress ?



OBJECTIVES

SPECIFIC OBJECTIVE

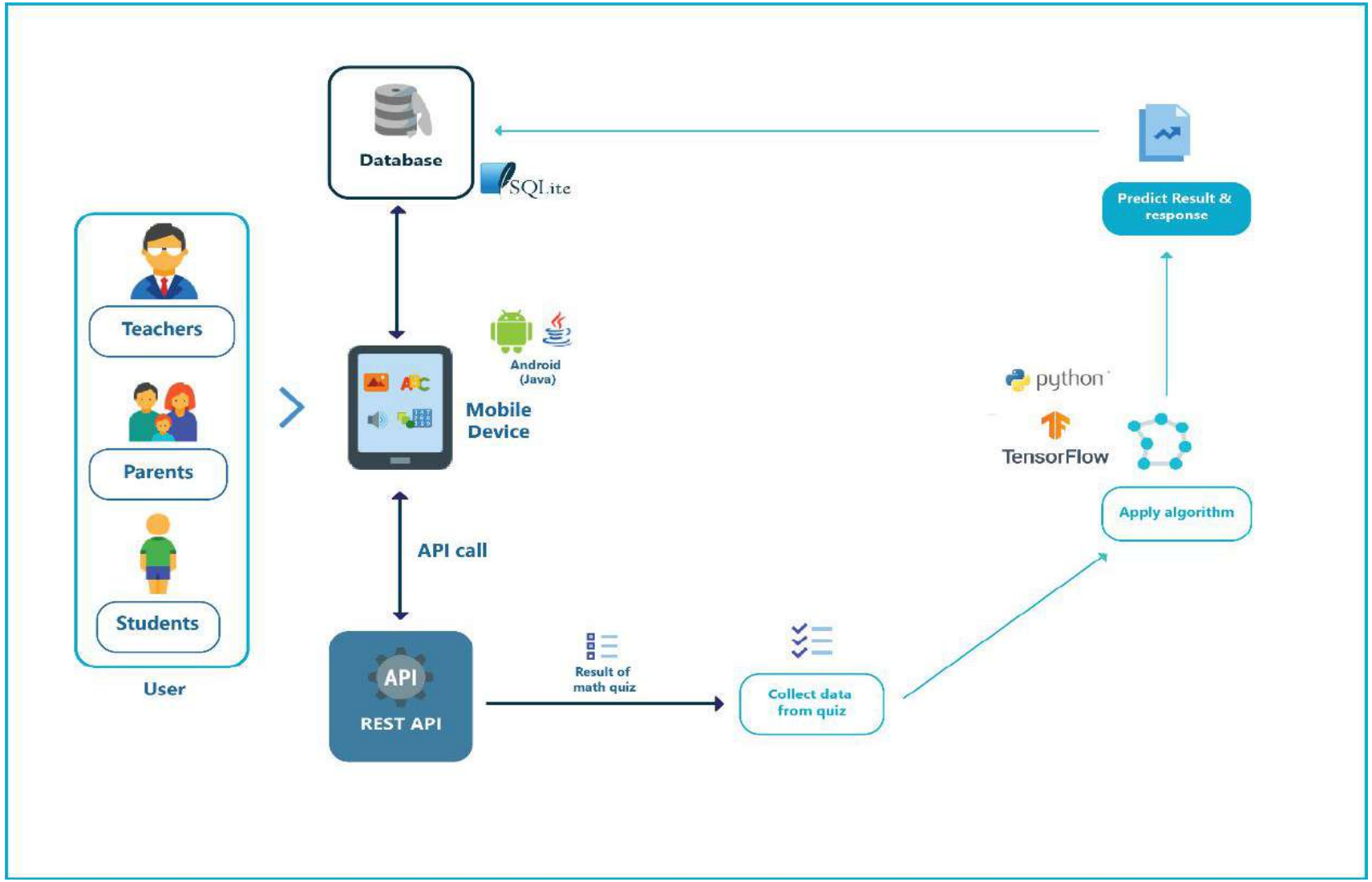
- Identify the risk for Operational and Practognostic Dyscalculia and provide activities to improve the condition.



Sub Objectives

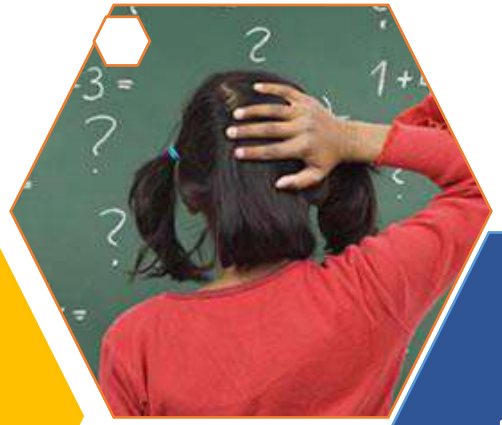
- Identify risk of Operational and Practognostic Dyscalculia.
- Provide activities for practice to improve the condition.
- Keep the track of student progress.
- Create Screening model

RESEARCH METHODOLOGY



Component Overview Diagram

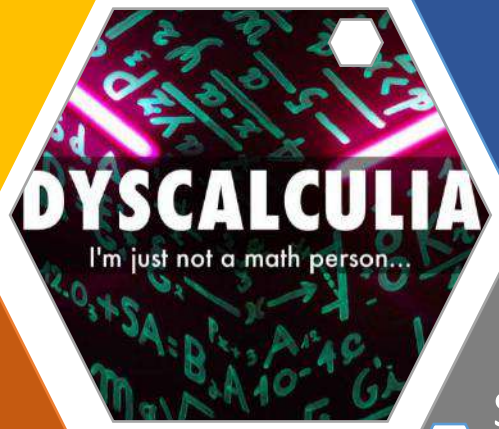
KEY PILLARS



Machine Learning

Data mining







Supervised Learning

Support Vector Machine

ACHIEVEMENTS



- Data Collection for Model Training
- Screening part of the Identify risk of Operational and Practognostic Dyscalculia (Created an interactive math quiz)
- Activities for practice to improve the condition (remedies).
- Keep track Student progress.
- ML model has been trained with 98% Accuracy.
- Implemented Backend Endpoints with Python Flask server to make RESTful API call.

ACHIEVEMENT EVIDENCE



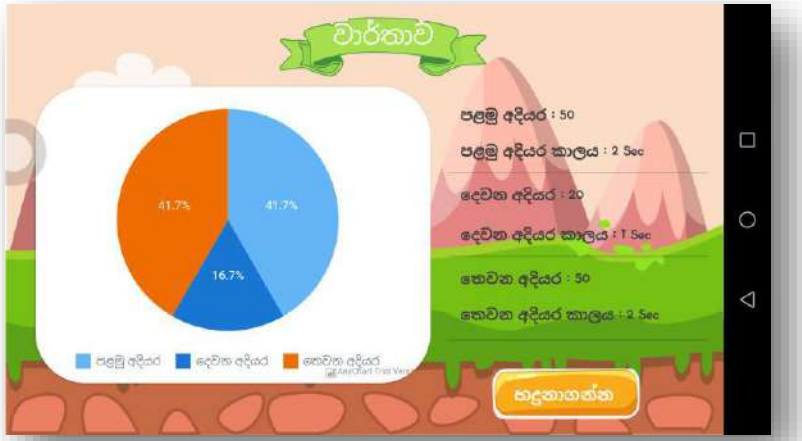
Stage 1



Stage 2



Stage 3



Result



Report



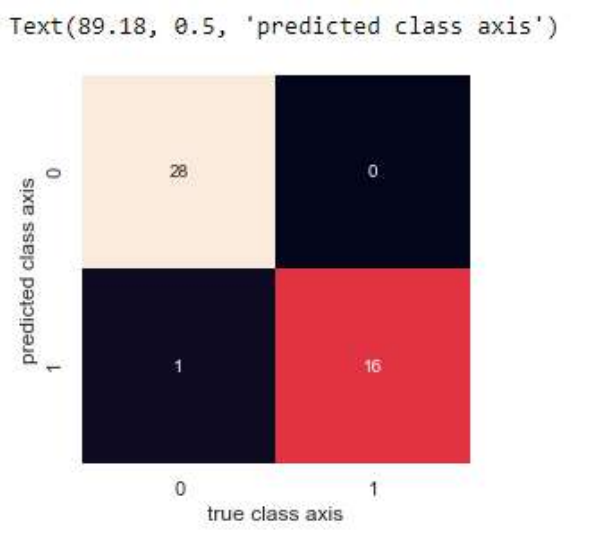
Remedy Activities

ACHIEVEMENT EVIDENCE CONT.

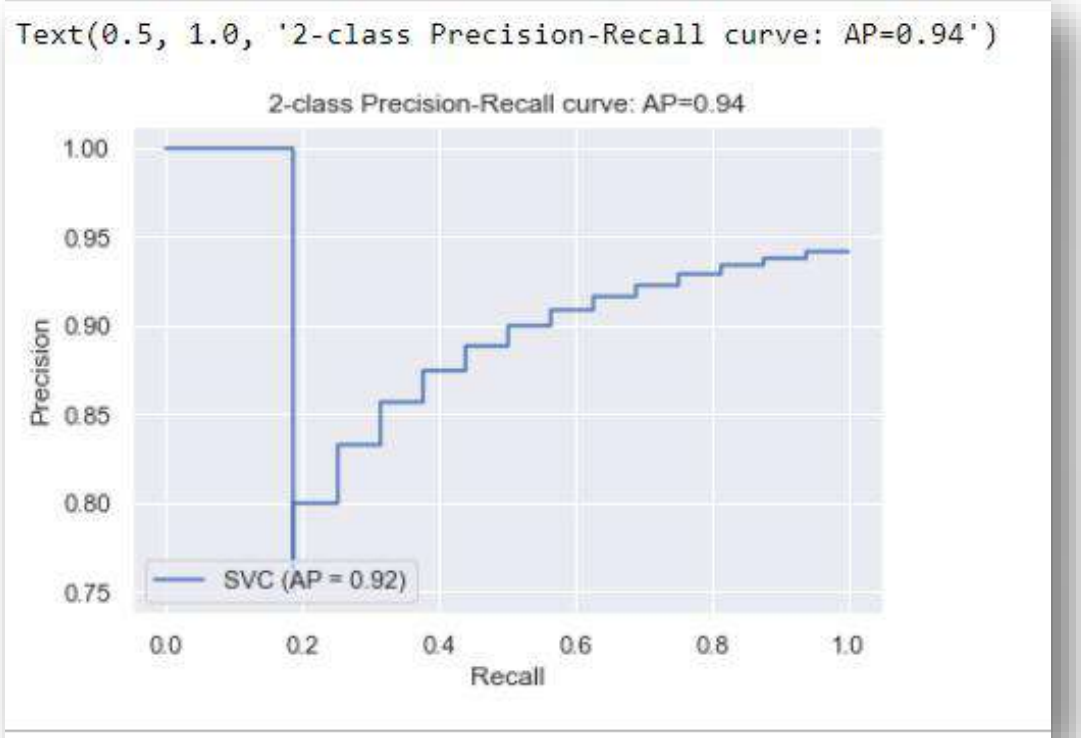
Classification Report :

	precision	recall	f1-score	support
No Dyscalculia	0.97	1.00	0.98	28
Dyscalculia	1.00	0.94	0.97	17
accuracy			0.98	45
macro avg	0.98	0.97	0.98	45
weighted avg	0.98	0.98	0.98	45

Classification report



Test Prediction Result



Precision recall Curve for Dyscalculia Screening

WHAT IS TO BE DONE ?



- Finishing the UI/UX design
- Deploy the Backend to the Cloud Server
- Integrate with Backend and Frontend

❖ REFERENCES

[1] Dysgraphia Information Page. (n.d.). Retrieved July 21, 2020, from [Online]. Available:https://www.ninds.nih.gov/Disorders/All-Disorders/Dysgraphia-Information-Page [Accessed: 26-July- 2020]

[2] Equal rights to education for children with Learning Disabilities in Sri Lanka [Online]. Available: http://www.unicef.lk/everymind/ [Accessed: 26-July-2020]

[3] N. C. Zygouris *et al.*, "Screening for disorders of mathematics via a web application," *IEEE Glob. Eng. Educ. Conf. EDUCON*, no. April, pp. 502–507, 2017, doi: 10.1109/EDUCON.2017.7942893.

[4] O. Poobrasert and W. Gestubtim, "Development of assistive technology for students with dyscalculia," *2013 2nd Int. Conf. E-Learning E-Technologies Educ. ICEEE 2013*, no. i, pp. 60–63, 2013, doi: 10.1109/ICeLeTE.2013.6644348.

[5] Universiti Tenaga Nasional, Universiti Tenaga Nasional. College of Information Technology, IEEE Computer Society, IEEE Malaysia Section, and Institute of Electrical and Electronics Engineers, "Conference Proceedings - 6th International Conference on Information Technology and Multimedia at UNITEN: Cultivating Creativity and Enabling Technology Through the Internet of Things, ICIMU 2014," *Conference Proceedings - 6th International Conference on Information Technology and Multimedia at UNITEN: Cultivating Creativity and Enabling Technology Through the Internet of Things, ICIMU 2014*. 2015.

[6] N. Beacham and C. Trott, "Screening for Dyscalculia within HE," *MSOR Connect.*, vol. 5, no. 1, Feb. 2005, doi: 10.11120/msor.2005.05010004.

[7] M. S. Margaj, "Significance of Data Mining Techniques in Classifying Dyscalculia," *Int. J. Eng. Comput. Sci.*, vol. 5, no. 08, pp. 1–6, 2016, doi: 10.18535/ijecs/v5i8.62.



IT17186766 | Prabath Shalitha Liyanage

BSc (Hons) Degree in Information Technology (specialization in Software Engineering)

THE PROGRESS

Identify the risk for Lexical and Verbal Dyscalculia and provide activities to improve the condition

INTRODUCTION

BACKGROUND

- Dyscalculia is a biologically based learning disability related to mathematic, numbers, and arithmetic skills.
- Centered around Neurological system.
- It is generally seen as mathematical equivalent of Dyslexia.
- According to British Dyslexia Association, Dyscalculia occur in 3 - 6% of population [1].
- There are 5 main types of Dyscalculia [2].
- Two types of them are focusing on here,

1. **Verbal Dyscalculia**
2. **Lexical Dyscalculia**



RESEARCH GAP

Features	1	2	3	4	Nena Shilpa
Number identification	✓	✓	✓	✓	✓
Voice recognition techniques	✗	✗	✗	✗	✓
Support Sinhala Language	✗	✗	✗	✗	✓
Improved UI/UX	✓	✓	✗	✓	✓
Keep track of progress	✗	✗	✗	✗	✓
Report Generation	✗	✓	✗	✗	✓
ML techniques	✗	✗	✗	✗	✓





1 - Mobile App for Dyscalculia Children in Malaysia [3]

2 - Mobile App Development on Detection and Diagnose of Learning Disability for Children [4]

3 - Intelligent System for Diagnosing Learning Disorders in Children [5]

4 - KidKanit

RESEARCH PROBLEM

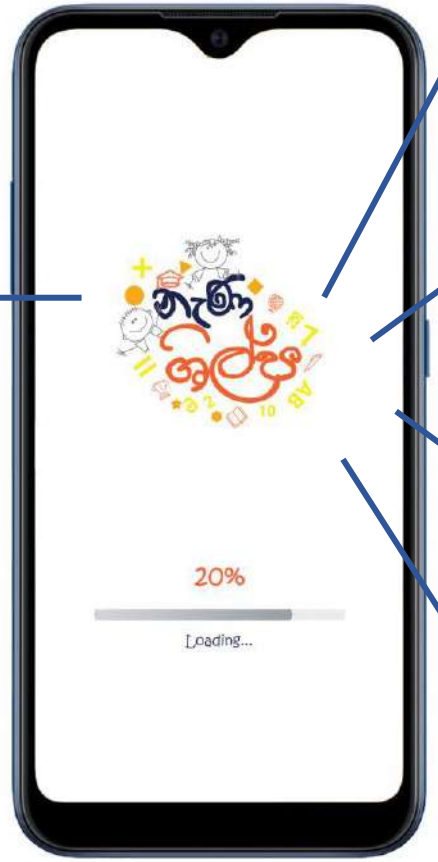


- How to Identify the risk for Lexical and Verbal Dyscalculia ?
- How to Provide activities to improve the condition ?
- How to Keep the track of progress ?

OBJECTIVES

Specific Objective

Identify the risk of Lexical and Verbal Dyscalculia and provide activities to improve the condition

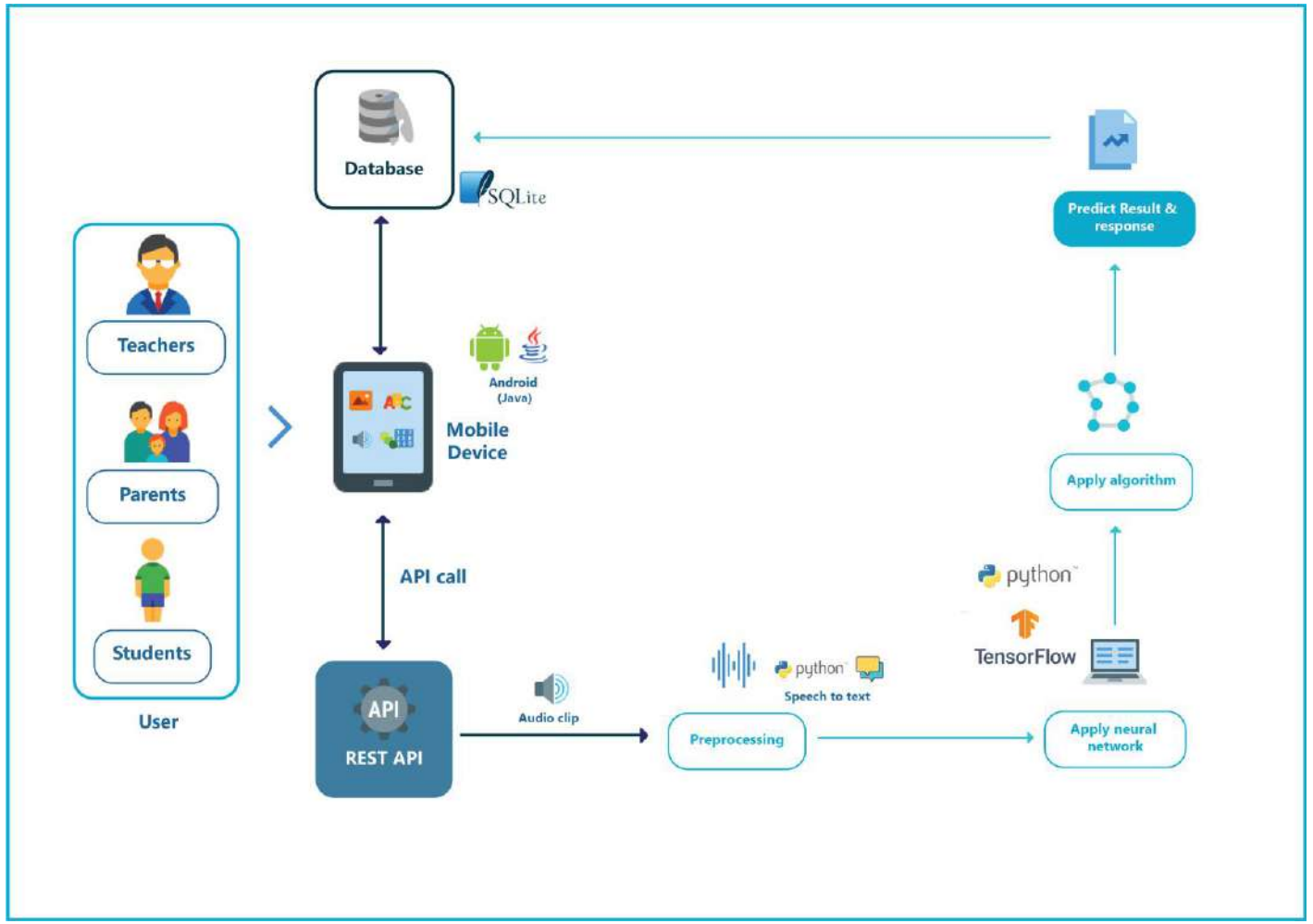


Sub Objectives

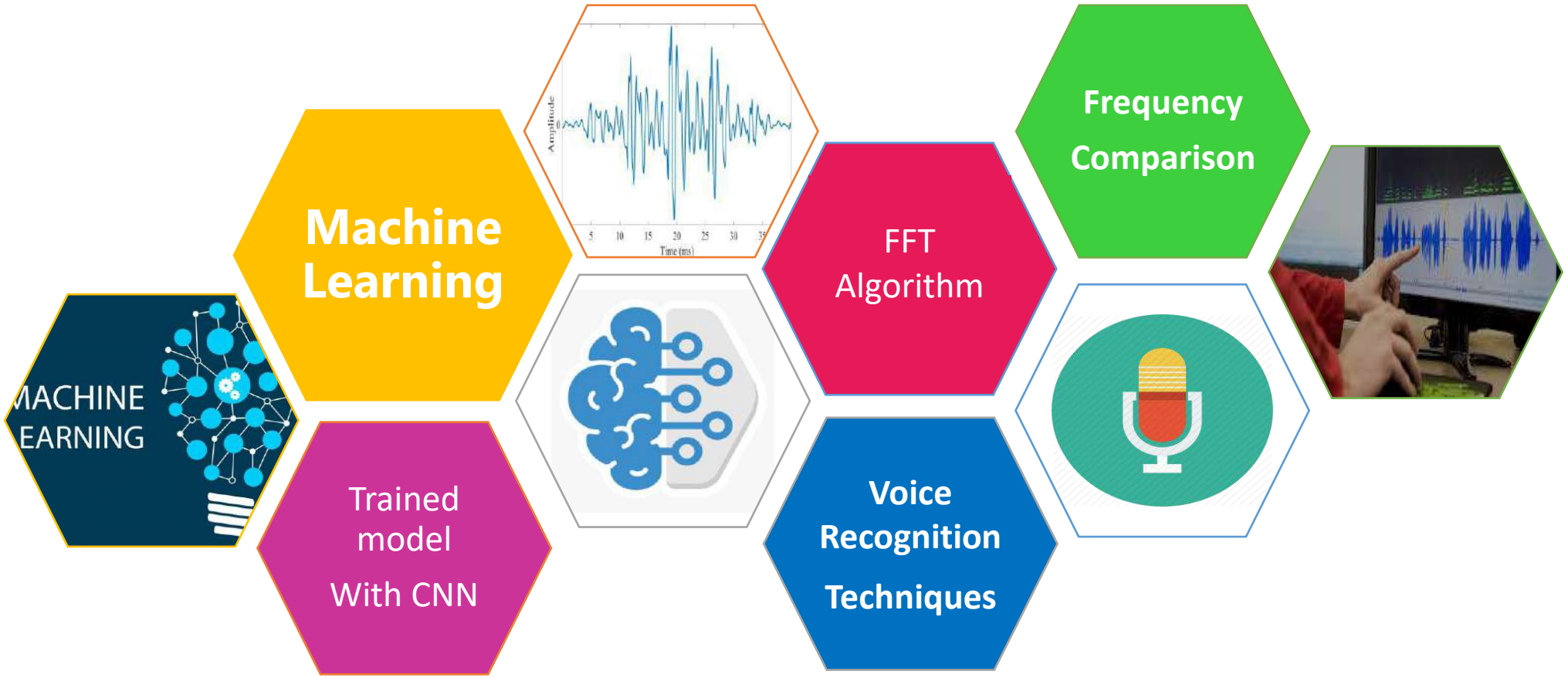
- Identify risk of Lexical Dyscalculia.
- Identify risk of Verbal Dyscalculia.
- Generate report with performed results
- Suggest remedies for identified user with risk.

METHODOLOGY

COMPONET OVERVIEW DIAGRAM



KEY PILLARS


Machine Learning
MACHINE EARNING
Trained model With CNN
Amplitude
Time (ms)
FFT Algorithm
Voice Recognition Techniques
Frequency Comparison

ACHIEVEMENTS



- Data gathering.
- UI/UX designe.
- Tests for identify Verbal & Lexical Dyscalculia.
- Practice remedies.
- Save test records.
- 91% CNN model accuracy.

ACHIEVEMENT EVIDENCES



Sub Menu



Test Select



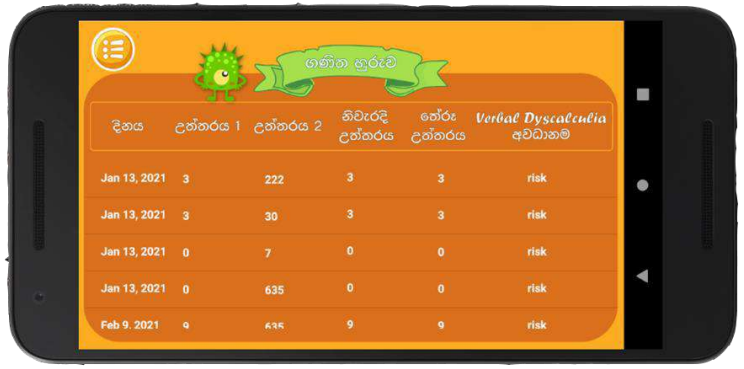
Test Type 1



Test Type 2



Remedy Activity

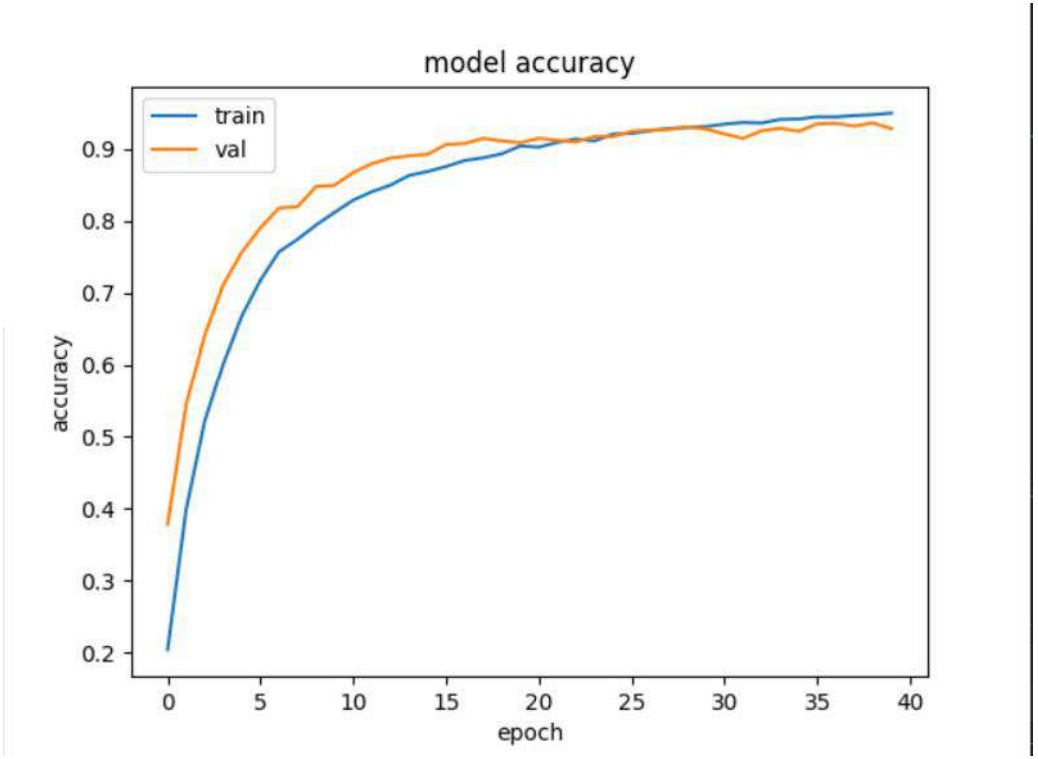


Previous Records

ACHIEVEMENT EVIDENCES

```
Model: "sequential"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
conv2d (Conv2D)              (None, 42, 11, 64)        640
_________________________________________________________________
batch_normalization (BatchNo (None, 42, 11, 64)        256
_________________________________________________________________
max_pooling2d (MaxPooling2D) (None, 21, 6, 64)         0
_________________________________________________________________
conv2d_1 (Conv2D)            (None, 19, 4, 32)         18464
_________________________________________________________________
batch_normalization_1 (Batch (None, 19, 4, 32)         128
_________________________________________________________________
max_pooling2d_1 (MaxPooling2 (None, 10, 2, 32)         0
_________________________________________________________________
conv2d_2 (Conv2D)            (None, 9, 1, 32)          4128
_________________________________________________________________
batch_normalization_2 (Batch (None, 9, 1, 32)          128
_________________________________________________________________
max_pooling2d_2 (MaxPooling2 (None, 5, 1, 32)          0
_________________________________________________________________
flatten (Flatten)            (None, 160)               0
_________________________________________________________________
dense (Dense)                (None, 64)                10304
_________________________________________________________________
dropout (Dropout)            (None, 64)                0
_________________________________________________________________
dense_1 (Dense)              (None, 10)                650
=================================================================
Total params: 34,698
Trainable params: 34,442
Non-trainable params: 256
_________________________________________________________________
```

```
Test error: 0.32437241077423096, test accuracy: 0.916939914226532
```



WHAT IS TO BE DONE

- Improve UI/UX.
- Deploy back-end to Cloud server.
- Apply noise Reduction technics for captured audios.
- Integrate back-end with front-end.



REFERENCES

[1] "Number Dyslexia: Everything to know about Dyscalculia - Number Dyslexia", *Number Dyslexia*, 2020. [Online]. Available: https://numberdyslexia.com/dyscalculia-aka-number-dyslexia/

[2] "CogniFit", *Beating dyscalculia: Treatment, Exercises, Causes, Symptoms, Types, Assessment and Definition*, 2020. [Online]. Available: https://www.cognifit.com/pathology/dyscalculia.

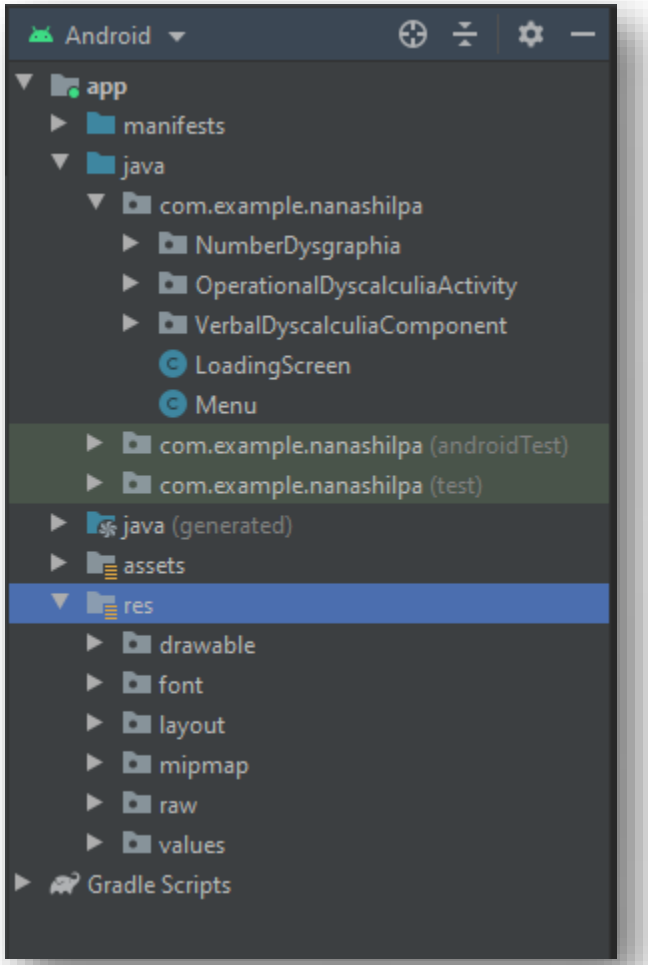
[3] M. Mohd Ariffin, F. A. Abd Halim, and N. Abd Aziz, "Mobile application for dyscalculia children in Malaysia," *Proc. 6Th Int. Conf. Comput. Informatics Embrac. Eco-Friendly Comput.*, no. April, pp. 467–472, 2017, [Online]. Available: http://www.uum.edu.my.

[4] G. Rajivsureshkumar, K. Malarvizhi, and G. Deebanchakkarawarthi, "Mobile Application Development on Detection and Diagnose of Learning Disability for Children," no. 6, pp. 216–224, 2019.

[5] M. A. J. Member and B. Shilpa, "Intelligent System for Diagnosing Learning Disorders in Children," *Lect. Notes Eng. Comput. Sci.*, vol. 2186, no. 1, pp. 70–75, 2010.



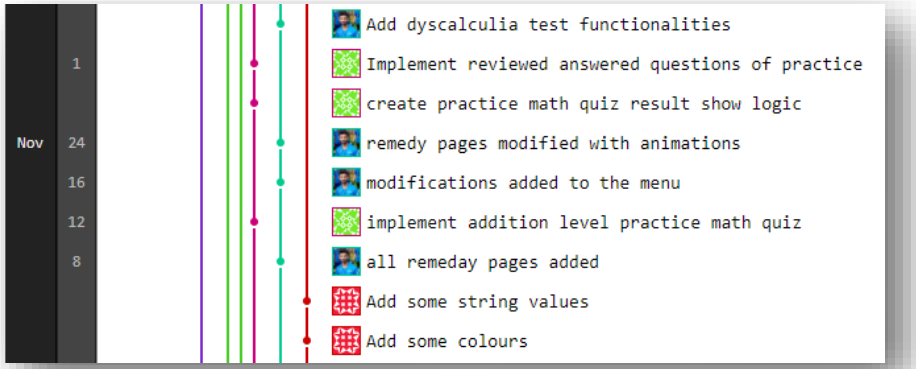
7. STANDARDS AND BEST PRACTICES



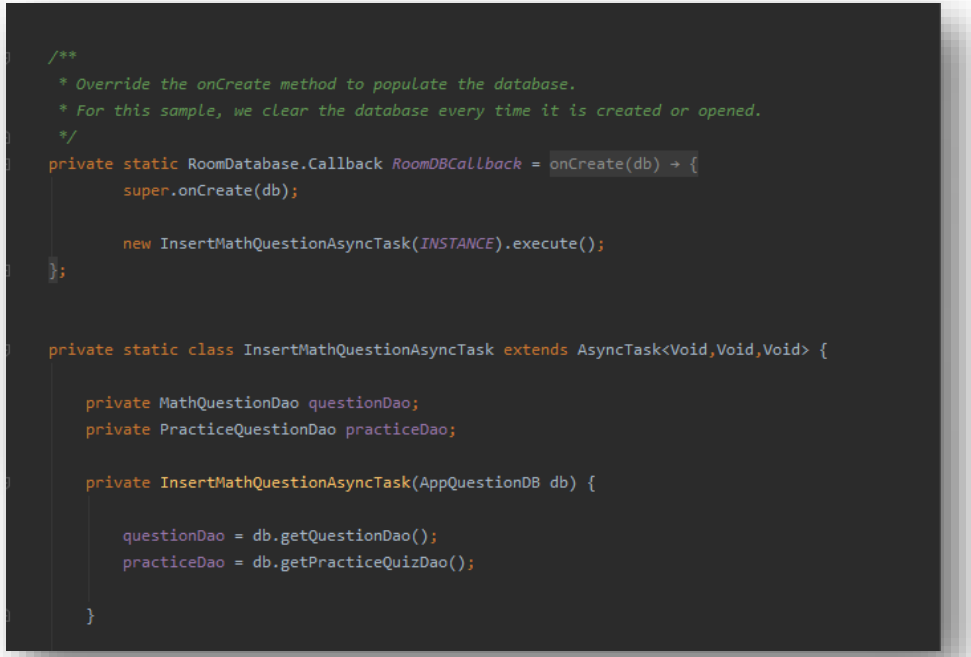
Packages for Component Organization



Industry standard for project architecture



Descriptive Source control



```
/**
 * Override the onCreate method to populate the database.
 * For this sample, we clear the database every time it is created or opened.
 */
private static RoomDatabase.Callback RoomDBCallback = onCreate(db) → {
        super.onCreate(db);

        new InsertMathQuestionAsyncTask(INSTANCE).execute();
};


private static class InsertMathQuestionAsyncTask extends AsyncTask<Void,Void,Void> {

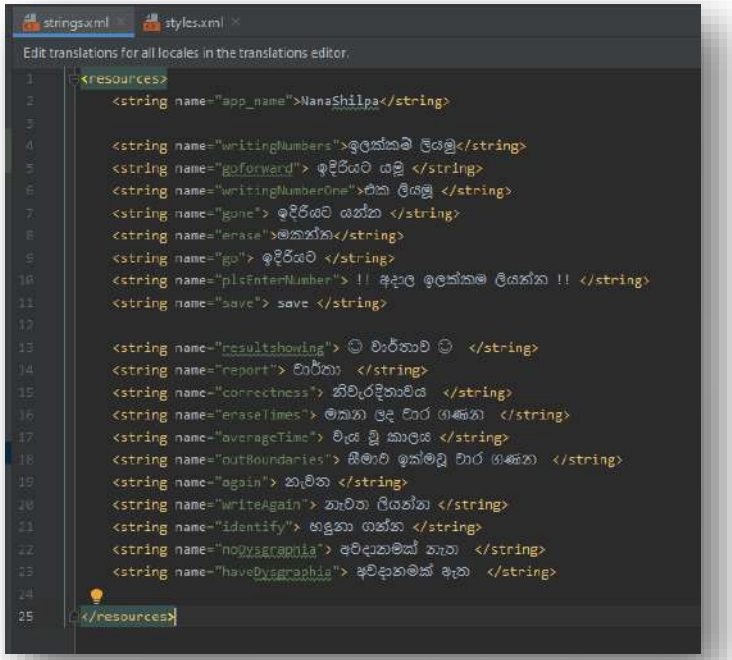
    private MathQuestionDao questionDao;
    private PracticeQuestionDao practiceDao;

    private InsertMathQuestionAsyncTask(AppQuestionDB db) {

        questionDao = db.getQuestionDao();
        practiceDao = db.getPracticeQuizDao();

    }
```

Increase code readability using developer comment



```
stringsxml  stylesxml
Edit translations for all locales in the translations editor.
<resources>
    <string name="app_name">NanaShilpa</string>

    <string name="writingNumbers">ඉලක්කම් ලියමු</string>
    <string name="goforword"> ඉදිරියට යමු </string>
    <string name="writingNumberOne">එක ලියමු </string>
    <string name="gone"> ඉදිරියට යන්න </string>
    <string name="erase">මකන්න</string>
    <string name="go"> ඉදිරියට </string>
    <string name="plsEnterNumber"> !! අදාල ඉලක්කම ලියන්න !! </string>
    <string name="save"> save </string>

    <string name="resultshowing"> ඔබ වැර්තාව ඔබ  </string>
    <string name="report"> වාර්තා  </string>
    <string name="correctness"> නිවැරදිතාවය  </string>
    <string name="eraseTimes"> මකන්න ලද වාර ගණන  </string>
    <string name="averageTime"> වැය වූ කාලය </string>
    <string name="outBoundaries"> සීමාව ඉක්මවූ වාර ගණන  </string>
    <string name="again"> නැවත </string>
    <string name="writeAgain"> නැවත ලියන්න </string>
    <string name="identify"> හඳුනා ගන්න </string>
    <string name="noDysgraphia"> අවිදානමක් නැත  </string>
    <string name="haveDysgraphia"> අවිදානමක් ඇත  </string>

</resources>
```

Resource file for better maintainability

8. REQUIREMENTS

Personal Requirements

1. Students
2. Teachers
3. Parents
4. Medical officer



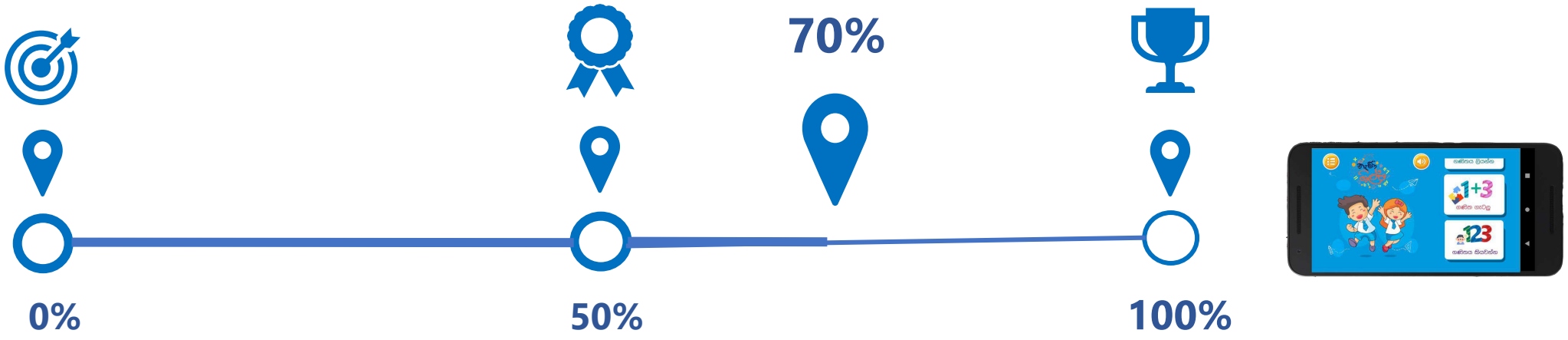
System Requirements

- ❑Software
  - **User-end**
    - ➢Android version 5 or above
  - **Developer-end**
    - ➢Android studio
    - ➢TensorFlow v2.0
    - ➢Python v3.7
    - ➢SQLite
- ❑Hardware
  - ➢Speaker
  - ➢ Microphones
  - ➢User Mobile Touch Screen

Non-functional Requirements

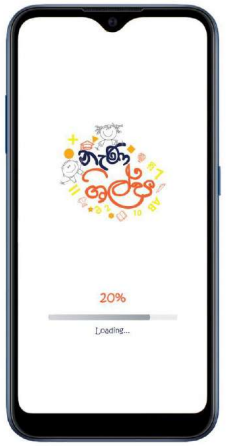
- ➢Reliability
- ➢Security
- ➢Performance
- ➢Usability
- ➢Availability

ADDRESSING THE FUNCTIONAL REQUIREMENT IN IMPLEMENTATION



- For the currently completed component implementation, all the functional requirements have been addressed which we had previously mentioned on the proposal presentation.

9. COMMERCIALIZATION ASPECTS



Using the proposed application, we hope to provide low-cost subscription plans considering each side.



For Schools
Rs. 1500

Permanent Subscription
Rs. 1000

Monthly Subscription
Rs. 300

Annual Subscription
Rs. 600

Low income Families.
Free Subscription.

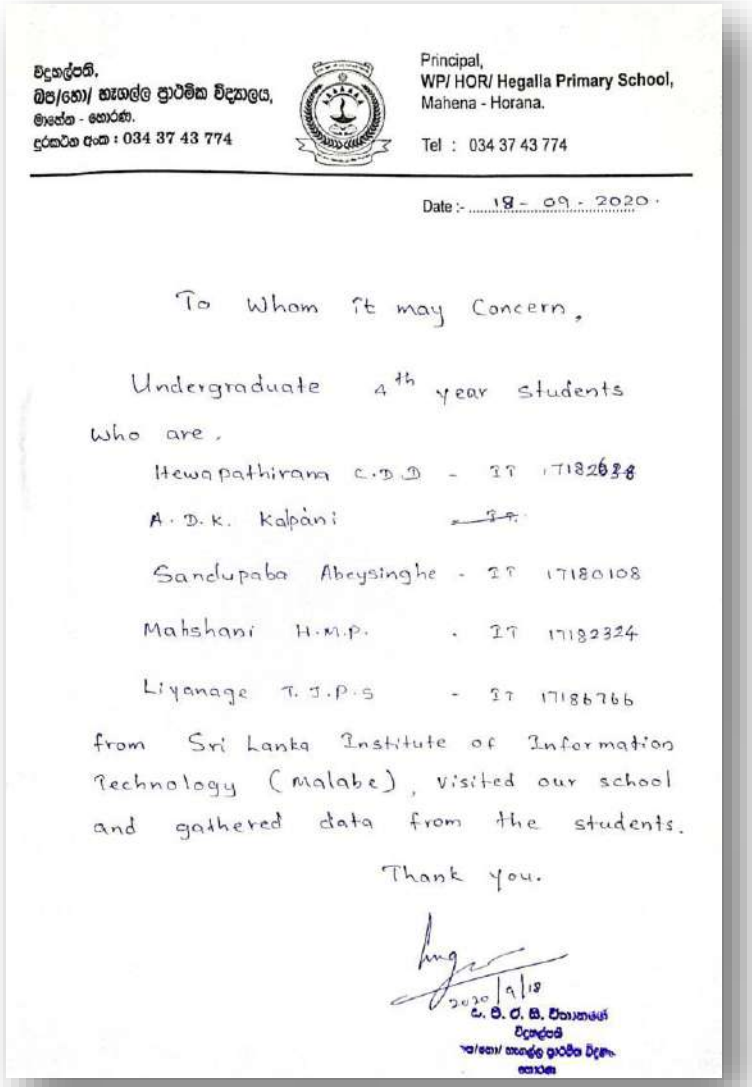
Child Orphanages.
Free Subscription

Social Media Commercialization.

10. BUDGET

Component	Amount (USD)	Amount (LKR)
Traveling fee for the possible consultation sessions and data gathering (Malabe, School selected for data extracting)	1.08	200.00
Internet charges (the development and technical information learning)	10.77	2000.00
Technical consultation charges (External technical information session)	5.4	1000.00
Total	17.25	3200.00

APPENDIX



විදුහල්පති,
බප/හෙ/ හැඟල්ල ප්‍රාථමික විද්‍යාලය,
මාහේන - හොරණ.
දුරකථන අංක : 034 37 43 774

Principal,
WP/ HOR/ Hegalla Primary School,
Mahena - Horana.
Tel : 034 37 43 774

Date :- 18 - 09 - 2020

To Whom it may Concern,

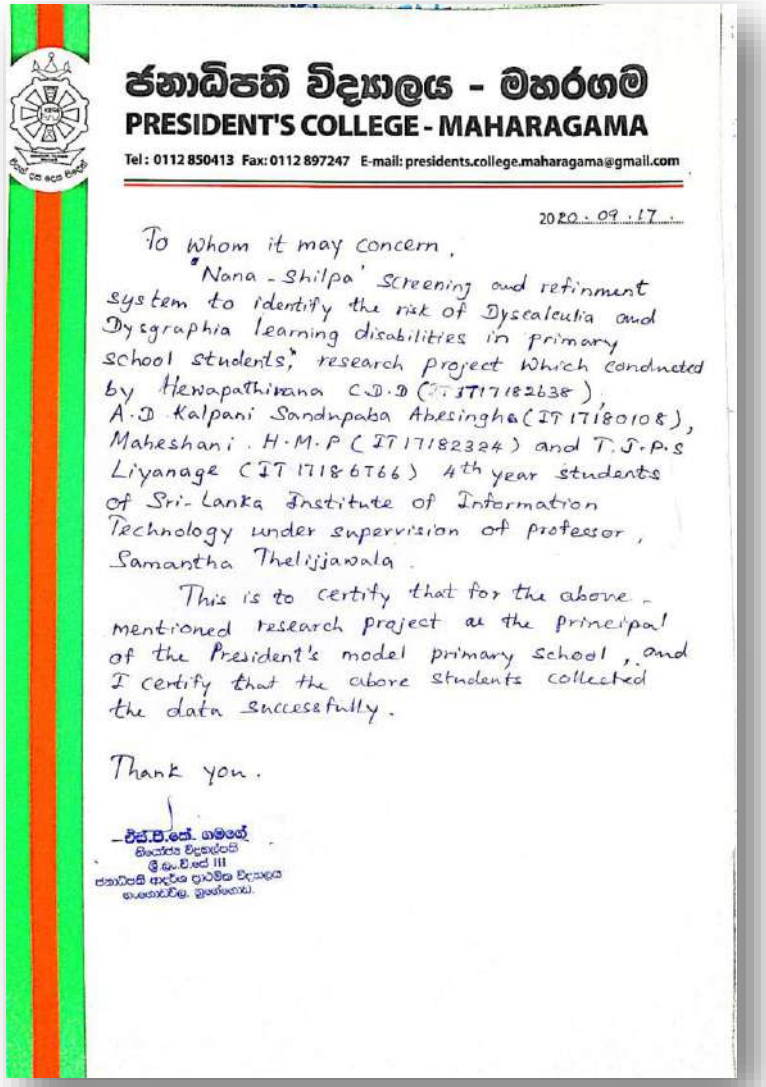
Undergraduate 4th year students who are,

Hewapathirana C.D.D - IT 17182638
A.D.K. Kalpani
Sandupaba Abeysinghe - IT 17180108
Mahshani H.M.P. - IT 17182324
Liyanage T.J.P.S - IT 17186766

from Sri Lanka Institute of Information Technology (Malabe), visited our school and gathered data from the students.

Thank you.

2020/9/18
ඒ. බී. ඩ. සී. වීරසේකර
විදුහල්පති
බප/හෙ/ හැඟල්ල ප්‍රාථමික විද්‍යා.
හොරණ



ජනාධිපති විද්‍යාලය - මහරගම
PRESIDENT'S COLLEGE - MAHARAGAMA
Tel : 0112 850413 Fax: 0112 897247 E-mail: presidents.college.maharagama@gmail.com

2020 . 09 . 17

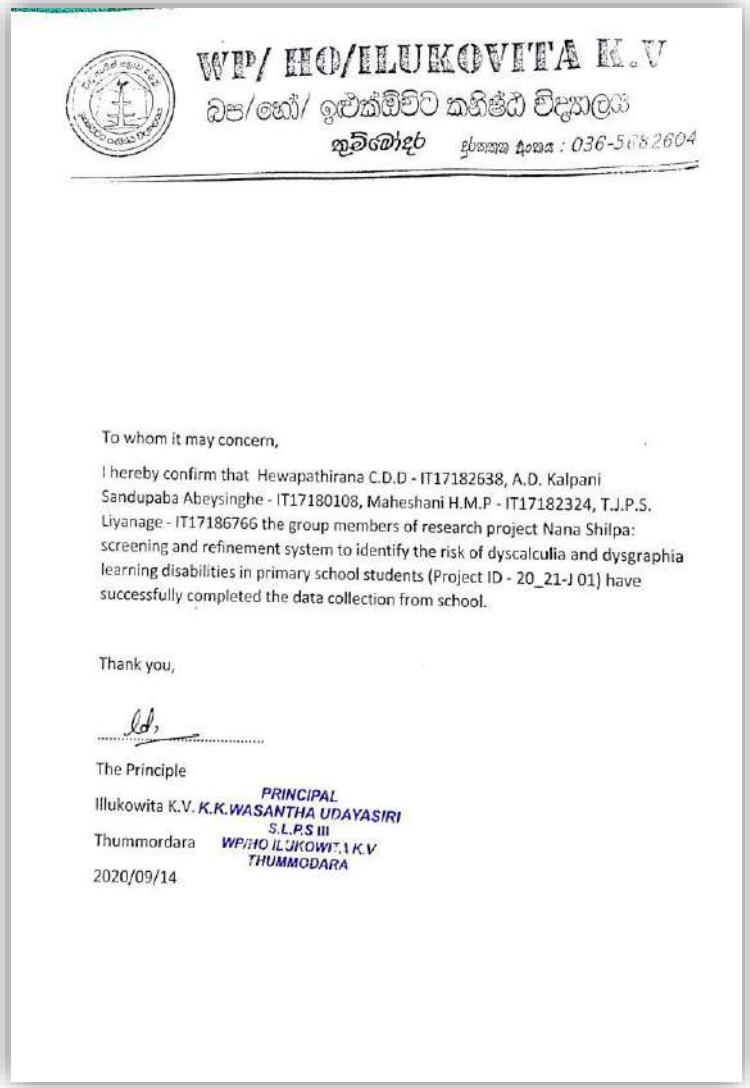
To whom it may concern,

'Nana - Shilpa' screening and refinment system to identify the risk of Dyscalculia and Dysgraphia learning disabilities in primary school students, research project which conducted by Hewapathirana C.D.D (IT 17182638), A.D Kalpani Sandupaba Abesingha (IT 17180108), Maheshani . H.M.P (IT 17182324) and T.J.P.S Liyanage (IT 17186766) 4th year students of Sri-Lanka Institute of Information Technology under supervision of professor, Samantha Thelijjawala.

This is to certify that for the above - mentioned research project at the principal of the President's model primary school, and I certify that the above students collected the data successfully.

Thank you.

ටී.ජී.ජී.කේ. ගමගේ
විදුහල්පති
ජූ.ඛ.විශේ III
ජනාධිපති ආදර්ශ ප්‍රාථමික විද්‍යාලය
මහරගම, ශ්‍රීලංකාව.



WP/ HO/ILUKOVITA K.V
බප/හෝ/ ඉළුක්ඔවිට කනිෂ්ඨ විද්‍යාලය
තුම්මොදර දුරකථන අංකය : 036-5682604

To whom it may concern,

I hereby confirm that Hewapathirana C.D.D - IT17182638, A.D. Kalpani Sandupaba Abeysinghe - IT17180108, Maheshani H.M.P - IT17182324, T.J.P.S. Liyanage - IT17186766 the group members of research project Nana Shilpa: screening and refinement system to identify the risk of dyscalculia and dysgraphia learning disabilities in primary school students (Project ID - 20_21-J 01) have successfully completed the data collection from school.

Thank you,

The Principle
Illukowita K.V. K.K.WASANTHA UDAYASIRI
S.L.P.S III
Thummordara WP/HO ILUKOWITA K.V
THUMMODARA
2020/09/14

APPENDIX





DEMOSTRATION







Q & A.



THANK YOU